



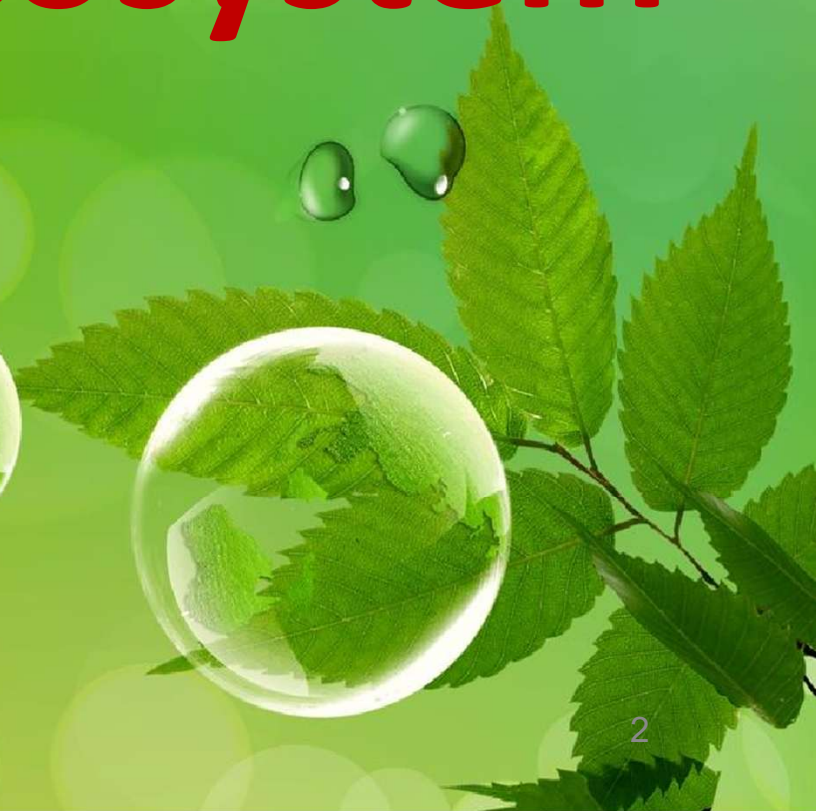
Department of Chemistry

Environmental Chemistry (2 credit)

CY1010

July 2026

Environment & Ecosystem



INSTRUCTORS

Dr. Anup Bhunia

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Dr. Abhijit Sau

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Dr. Kishore Natte

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Dr. M. Annadhasan

Email: annadhasan@chy.iith.ac.in



Course Content

Part 1 (Dr. Anup): Know our environment (earth-life supporting systems, chemistry of lithosphere, energy balance, sustainability, and recycle), know about global warming (infrared absorption, molecular vibration, atmospheric window, residence time of greenhouse gases, evidences and effects of global warming). Future challenges (CO₂ sequestering and nuclear energy).

Part 2 (Dr. Abhijit): Aspects of transformations in atmosphere (microbial degradation of organics-environmental degradation of polymers, atmospheric lifetime, toxicity). Green chemistry and industrial Ecology.

Part 3 (Dr. Kishore): Organic chemicals in the environment, insecticides, pesticides, herbicides and insect control, soaps, synthetic surfactants, polymers, and haloorganics. Fate of organic/inorganic chemicals in natural and engineered systems (fate of polymers after use, detergents, synthetic surfactants insecticides, pesticides etc. after use).

Part 4 (Dr. Annadhasan): Deeper analysis of atmospheric pollution (chemistry of CO, NO_x, VOCs, SO₂, industrial smog, photochemical smog), ozone depletion (production, catalytic destruction)

Key topics of CY1010 course

- Global warming/greenhouse gases
- Ozone depletion
- CO₂ capturing/utilization
- Biofuels
- Electrical and hydrogen fuel cell vehicles
- Plastic waste recycling
- Auto-exhaust purification
- Fate of polymers, detergents, synthetic surfactants, insecticides, pesticides etc. after use

Segment details

Segment 1-2

Part 1: Dr. Anup Bhunia

Part 2: Dr. Abhijit Sau

Segment 3-4

Part 3: Prof. Kishore Natte

Part 4: Dr. M. Annadhasan

Slot details

	9:00 9:55	10:00 10:55	11:00 11:55	12:00 12:55	12:55 14:30	14:30 15:55	16:00 17:25
Monday	A	B	C	D	Lunch	P	Q
	FN1					AN1	
Tuesday	D	E	F	G		R	S
	FN2					AN2	
Wednesday	B	C	A	G		F	Challenge Lectures
	FN3						
Thursday	C	A	B	E		Q	P
	FN4					AN4	
Friday	E	F	D	G		S	R
	FN5					AN5	

Segment	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	No. of Week Days	No. of Weekdays in Segment			
1	27 Jul	28 Jul	29 Jul	30 Jul	31 Jul		12	24			
	3 Aug	4 Aug	5 Aug	6 Aug	7 Aug						
	10 Aug	11 Aug									
2			12 Aug	13 Aug	14 Aug		12				
	17 Aug	18 Aug	19 Aug	20 Aug	21 Aug						
	24 Aug	25 Aug	Milad-Un-Nabi	27 Aug	28 Aug						
3	31 Aug	1 Sep	2 Sep	3 Sep	4 Sep		12	24			
	7 Sep	8 Sep	9 Sep	10 Sep	11 Sep						
	Ganesh Chathurdhi	15 Sep	16 Sep								
4				17 Sep	18 Sep		12		24		
	21 Sep	22 Sep	23 Sep	24 Sep	25 Sep						
	28 Sep	29 Sep	30 Sep	01 Oct	Gandhi Jayanthi						
	05 Oct										
5		06 Oct	07 Oct	08 Oct	09 Oct		12	24			
	12 Oct	13 Oct	14 Oct	15 Oct	16 Oct						
	Break	19 Oct	Dussehra	21 Oct	22 Oct				23 Oct		
	26 Oct	27 Oct	28 Oct				12				
6				29 Oct	30 Oct					12	
	2 Nov	3 Nov	4 Nov	5 Nov	6 Nov						
	9 Nov	10 Nov	11 Nov	12 Nov	13 Nov						

Segment	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	No. of Week Days	No. of Weekdays in Segment
1	27 Jul	28 Jul	29 Jul	30 Jul	31 Jul		12	24
	3 Aug	4 Aug	5 Aug	6 Aug	7 Aug			
	10 Aug	11 Aug						
2			12 Aug	13 Aug	14 Aug		12	
	17 Aug	18 Aug	19 Aug	20 Aug	21 Aug			
	24 Aug	25 Aug	Milad-Un-Nabi	27 Aug	28 Aug			
3	31 Aug	1 Sep	2 Sep	3 Sep	4 Sep		12	24
	7 Sep	8 Sep	9 Sep	10 Sep	11 Sep			
	Ganesh Chathurdhi	15 Sep	16 Sep					
4				17 Sep	18 Sep		12	
	21 Sep	22 Sep	23 Sep	24 Sep	25 Sep			
	28 Sep	29 Sep	30 Sep	01 Oct	Gandhi Jayanthi			
	05 Oct							
5		06 Oct	07 Oct	08 Oct	09 Oct		12	24
	12 Oct	13 Oct	14 Oct	15 Oct	16 Oct			
Break	19 Oct	Dussehra	21 Oct	22 Oct	23 Oct			
	26 Oct	27 Oct	28 Oct					
6				29 Oct	30 Oct		12	
	2 Nov	3 Nov	4 Nov	5 Nov	6 Nov			
	9 Nov	10 Nov	11 Nov	12 Nov	13 Nov			

Start

Part 1 and 2 – last instruction class

Exam-1

Part 3 and 4 starts here

Part 3 and 4 – last instruction class

Exam-2

Segment	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	No. of Week Days	No. of Weekdays in Segment	
1	28 Jul	29 Jul	30 Jul	31 Jul	01 Aug		12	24	
	04 Aug	05 Aug	06 Aug	07 Aug	08 Aug				
	11 Aug	12 Aug							
2			13 Aug	14 Aug	Independence Day		12		
	18 Aug	19 Aug	20 Aug	21 Aug	22 Aug				
	25 Aug	26 Aug	Ganesh Chaturthi	28 Aug	29 Aug				
	01 Sep								
3		02 Sep	03 Sep	04 Sep	* Id-ul-Milad		12		
	08 Sep	09 Sep	10 Sep	11 Sep	12 Sep				
	15 Sep	16 Sep	17 Sep	18 Sep					
4					19 Sep		12	24	
	22 Sep	23 Sep	24 Sep	25 Sep	26 Sep				
Break	29 Sep	30 Sep	01 Oct	Dussehra/ Mahatma Gandhi's Birthday	03 Oct				
4	06 Oct	07 Oct	08 Oct	09 Oct	10 Oct				
	13 Oct	14 Oct	15 Oct	16 Oct	17 Oct				

Part 3 and starts here

Exam-2

Start

Part 1 and 2 –
last instruction
class

Exam-1

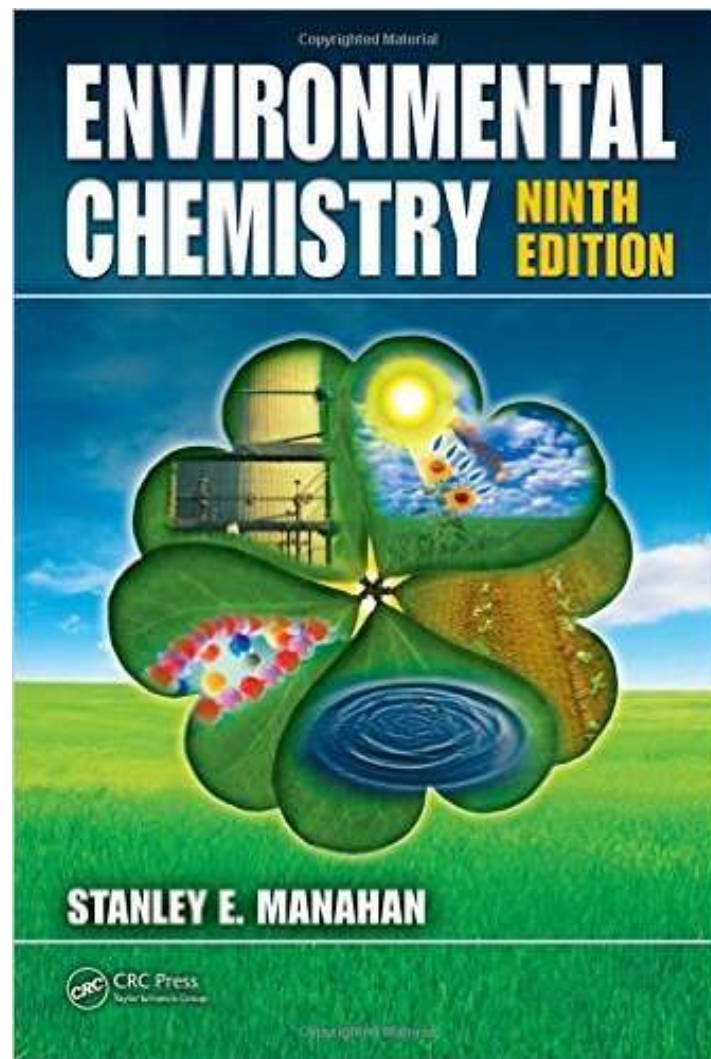
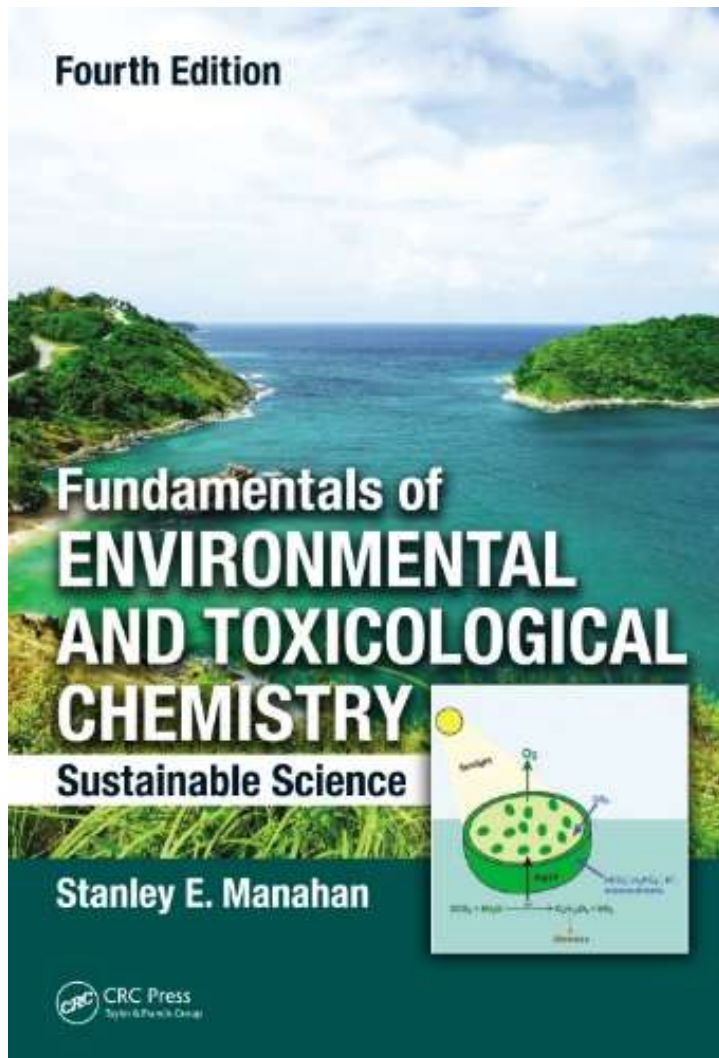
Part 3 and 4
starts here

Part 3 and 4 –
last instruction
class

Exam-2

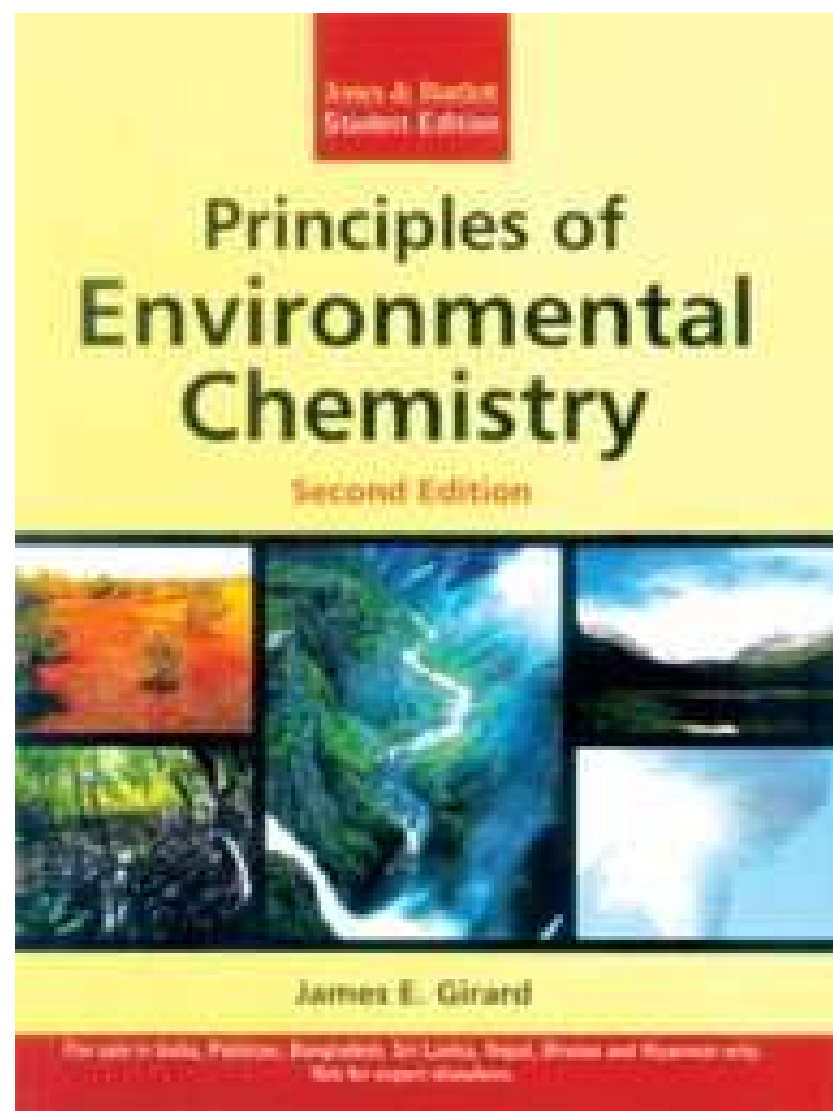
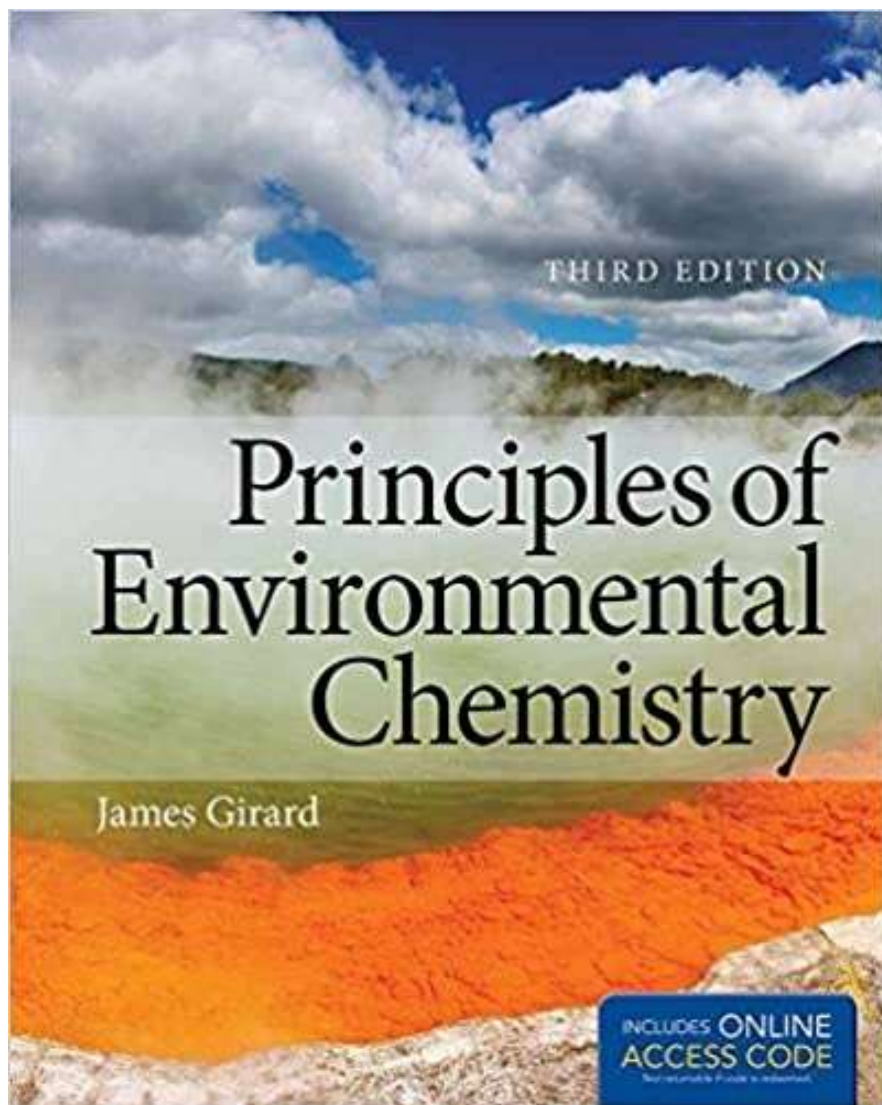
Environmental Chemistry

Reference books by Stanley



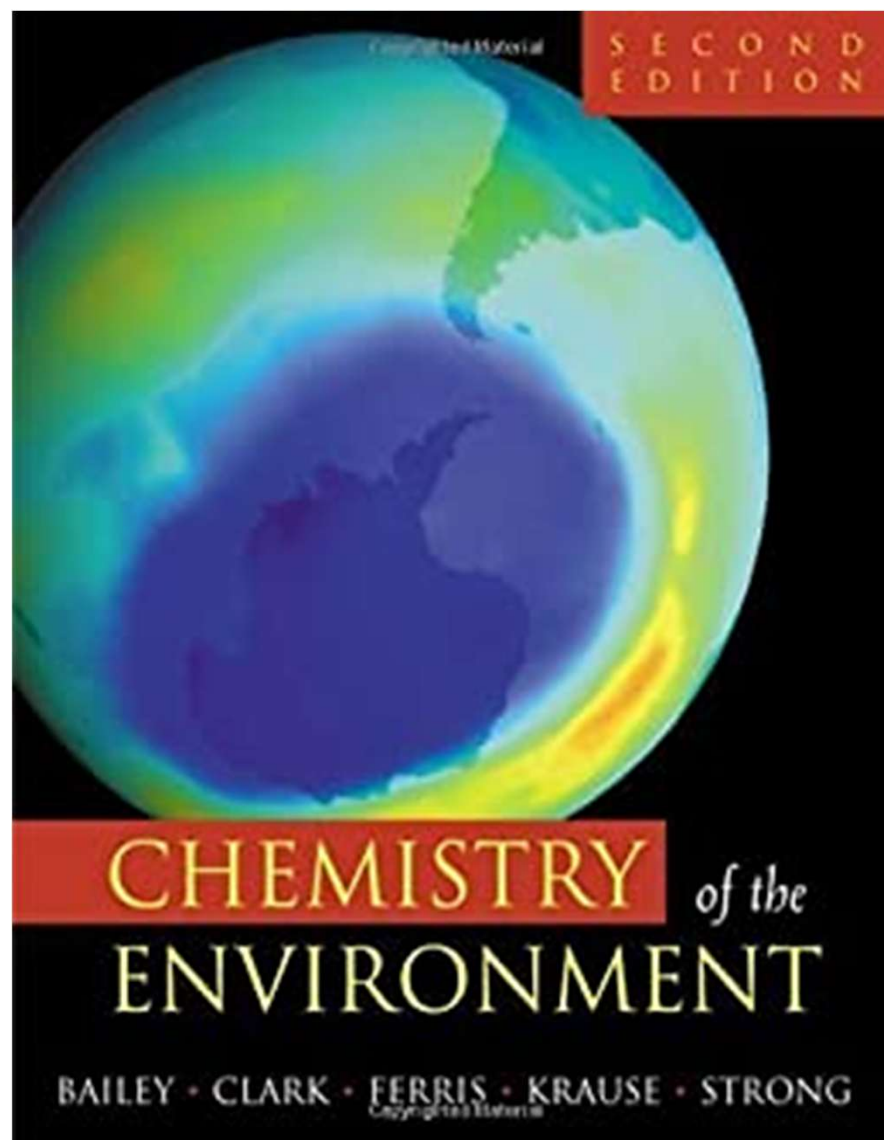
Principles of Environmental Chemistry

By James E. Girard



Chemistry of the Environment

by Bailey, Clark, Ferris, Krause, Strong



EXAM 1

SEGMENT 1

Syllabus from Dr. Anup and Dr. Abhijit teaching portions

Marks : 50

Type : MCQs

Date : 05th September 2026

Time : will be informed later

Venue : will be informed later

Duration : 90 min

EXAM 2

SEGMENT 2

Syllabus from Dr. Kishore and Dr. Annadhasan teaching portions

Marks : 50

Type : MCQs

Date : 10th October 2025

Time : will be informed later

Venue : will be informed later

Duration : 90 min

GRADING and EXAMS

SEGMENT 1-2

Exam 1 50 Marks

SEGMENT 3-4

Exam 2 50 Marks

Total

100 Marks

- **>95 : A+**
- **81-94 : A**
- **71-80 : A-**
- **61-70 : B**
- **51-60 : B-**
- **41-50 : C**
- **31-40 : C-**
- **21-30 : D**

A⁺, A 10

A⁻ 9

B 8

B⁻ 7

C 6

C⁻ 5

D 4